

Video Editing RL Mini-Benchmark

1 Benchmark Architecture

This benchmark evaluates long-horizon video-editing agents. Each run gives the agent a task prompt, source materials, output specifications, and a description of available editing tools. The agent does not see private ground truth, verifier thresholds, or other agents’ outputs. It works inside a Docker sandbox and submits a final video plus structured evidence: `submit/output.mp4`, `submit/edit_decision.json`, `submit/run_history.md`, and `submit/agent_transcript.md`. Some tasks also require sidecar artifacts such as captions.

Each task is released as one folder containing the prompt, source materials, verifier code entry points, private ground-truth criteria, and hackability analysis. The shared code exposes a single pipeline:

prepare workspace $\rightarrow$ run agent $\rightarrow$ verify submission $\rightarrow$ aggregate results.

The sandbox includes a CPU-based video and audio editing stack for media probing, trimming, concatenation, transcoding, frame extraction, audio processing, scene analysis, transcription, image analysis, schema validation, and report generation. Specific package names are listed in Appendix A.

The verifier is split into general and task-specific reward terms:

$$R = 0.2r_{\text{general}} + 0.8r_{\text{task}}.$$

r_{general} checks reusable media requirements: playable output, audio/video presence, format, duration gates, non-degenerate frames, audio quality, and required submitted files. r_{task} checks the actual editing objective. For tasks where semantic judgment is useful, the task reward is:

$$r_{\text{task}} = 0.85r_{\text{hard_task}} + 0.15r_{\text{llm}},$$

where $r_{\text{hard_task}}$ is explicit verifier code and r_{llm} is an LLM-as-judge score. For the A/V sync task, the LLM judge is diagnostic only; the final reward is hard-verifier based.

The reward design follows a “unit test” view of video editing. Instead of asking whether a video is good in one vague scalar, each dimension checks a concrete property: correct format, preserved content, removed artifacts, synced audio, readable captions, clean sound, or coherent source order. This gives denser RL signal than a single human preference label. The difficulty is reward hacking: a model can satisfy one narrow metric while hurting the video, such as making captions huge to pass readability. Each dimension therefore has counter-checks, and severe failures trigger hard caps so that many small passes cannot hide a major error.

2 Details of Benchmarks

2.1 Piecewise A/V Sync Repair

The agent repairs a damaged 16:9 tutorial clip. The source includes black boundary padding, four local A/V offsets, a white flash, a frozen/dead-air splice, a duplicate bad insert, and a local audio-level issue. The target horizon is about 15 genuinely sequential decisions and 28–34 tool calls. The

dependency structure is detection, timestamp localization, local repair planning, multi-pass rendering, and final verification. A one-command global shift is not enough because the sync error is piecewise.

The task stresses reasoning over a plan, tool use, and multimodal grounding. It does not intentionally stress iterative self-correction: agents can revise, but reward is based on the final diagnosis, edit, and media quality.

Issue	Location or magnitude	Expected repair
Boundary padding	1.2 s leader, 0.8 s tail	Remove padding without deleting useful content
Piecewise desync	+720, -650, +480, -380 ms regions	Repair local offsets, not one global shift
White flash	0.7 s visual artifact	Remove cleanly and preserve neighboring content
Freeze/dead air	0.75 s frozen splice	Remove dead section and maintain continuity
Duplicate insert	2.2 s repeated segment	Remove the duplicate only
Audio damage	24 s attenuated region	Normalize speech without clipping

Table 1: Injected issues for Piecewise A/V Sync Repair.

Reward rubric. The main task dimensions are black-boundary cleanup, global sync, segmented sync, freeze/dead-air cleanup, duplicate cleanup, visual-artifact cleanup, content preservation, source-time monotonicity, and audio-level repair. Sync and required-span preservation are hard-to-hack because they use private clean-reference timing and private required spans. Artifact and audio checks are hackable but bounded because over-trimming, silence, clipping, frozen frames, black frames, and duplicate content are measurable. LLM judging is soft/vibes and is not used as a primary reward for this task. The reward emits a vector of dense subscores plus caps for fatal media, bad duration, low content preservation, low task score, remaining major artifacts, duplicate failure, or severe audio failure. All rubric dimensions can enter the final training signal; with different weights they also provide denser, more step-level reward.

Agent	Model	r_{general}	r_{task}	Total
Codex	GPT-5.5	0.882	0.499	0.576
Codex	GPT-5.4	0.837	0.446	0.525
Codex	GPT-5.4 Mini	0.832	0.548	0.605
Claude Code	Opus 4.7	0.919	0.507	0.589
Claude Code	Sonnet 4.6	0.822	0.641	0.677
Claude Code	Haiku 4.5	0.832	0.432	0.512
Gemini CLI	3.1 Pro Preview	0.717	0.612	0.633
Gemini CLI	3.1 Flash-Lite	0.762	0.399	0.471

Table 2: Piecewise A/V Sync Repair results.

2.2 Expert Pancake Vertical Short

The agent converts a noisy pancake challenge/tutorial source into a vertical short. The realistic user request is to extract the useful expert cooking arc and remove novice, blooper, reaction, or end-card material. The output must be true portrait video, preserve the cooking sequence, include readable step captions, and avoid fake vertical formatting. The horizon is roughly 12–18 sequential decisions:

inspect source, identify tutorial regions, reject negative regions, plan step order, choose vertical crops, write captions, render, and verify mobile readability.

The task stresses planning, tool use, and visual grounding over action stages such as pan prep, batter, bubble cue, flip, and plating. It does not test generative video, object removal, or cinematic color transfer.

Reward rubric. Task dimensions include source authenticity, allowed-region coverage, negative-region leakage, visual step coverage, step order, vertical reframe quality, fake-portrait detection, caption completeness, caption/action alignment, and LLM-assessed tutorial coherence. Source-region and step-coverage checks are hard-to-hack because they use private intervals and source-frame matching. Caption and crop checks are hackable but bounded: the verifier checks caption size, placement, readability, important-object occlusion, and padding artifacts, so oversized or decorative captions do not pass for free. The LLM judge is soft/vibes and only a minority semantic layer. The reward is a vector of graded scores with caps for missing output, wrong aspect, fake vertical format, too much negative material, weak step coverage, order failure, missing captions, or weak semantic captions. All rubric dimensions can enter the final training signal; with different weights they also provide denser, more step-level reward.

Agent	Model	r_{general}	r_{task}	Total
Codex	GPT-5.5	0.887	0.796	0.805
Codex	GPT-5.4	0.948	0.961	0.939
Codex	GPT-5.4 Mini	0.881	0.879	0.865
Claude Code	Opus 4.7	0.928	0.809	0.808
Claude Code	Sonnet 4.6	0.896	0.749	0.767
Claude Code	Haiku 4.5	0.834	0.265	0.366
Gemini CLI	3.1 Pro Preview	0.897	0.651	0.350
Gemini CLI	3.1 Flash-Lite	0.892	0.712	0.550

Table 3: Expert Pancake Vertical Short results.

2.3 Rough Interview Caption Cleanup

The agent cleans up a rough interview clip into a publishable captioned segment. The task includes dead air, filler, uneven pacing, caption requirements, and semantic preservation constraints. A good edit removes obvious roughness but keeps the explanation intact and synchronized. The expected horizon is about 12–16 sequential decisions: inspect speech and timing, find removable dead regions, preserve semantic anchors, normalize audio, generate or repair captions, burn or export captions, and verify timing and layout.

This task stresses planning, tool use, audio-language grounding, and caption/video grounding. It does not test full dialogue rewriting, lip-sync generation, or synthetic narration.

Reward rubric. Task dimensions include pause cleanup, filler/dead-air reduction, semantic anchor recall, anchor order, source fidelity, caption file validity, ASR token F1, timing-aware token F1, caption-speech consistency, caption layout, duplicate text rate, and LLM-assessed publishability. Semantic anchors and source fidelity are hard-to-hack because they depend on private transcript anchors and source matching. Caption timing, audio cleanup, and readability are hackable but bounded because over-deletion, muting, clipping, oversized captions, subject occlusion, and desynchronization are checked. LLM quality is soft/vibes and used as a minority signal paired with hard caps. The verifier emits a dense subscore vector plus caps for fatal media, low semantic recall, weak pause removal,

missing captions, low source match, weak caption layout, low LLM quality, or obvious non-source content. All rubric dimensions can enter the final training signal; with different weights they also provide denser, more step-level reward.

Agent	Model	r_{general}	r_{task}	Total
Codex	GPT-5.5	0.816	0.787	0.792
Codex	GPT-5.4	0.745	0.798	0.787
Codex	GPT-5.4 Mini	0.809	0.667	0.696
Claude Code	Opus 4.7	0.824	0.664	0.696
Claude Code	Sonnet 4.6	0.785	0.728	0.739
Claude Code	Haiku 4.5	0.796	0.645	0.550
Gemini CLI	3.1 Pro Preview	0.831	0.838	0.837
Gemini CLI	3.1 Flash-Lite	0.733	0.630	0.651

Table 4: Rough Interview Caption Cleanup results.

3 Metric for Data Selection

To check whether the tasks measure distinct abilities, I form one reward vector per task across the same ordered model set:

(GPT-5.5, GPT-5.4, GPT-5.4 Mini, Opus, Sonnet, Haiku, Gemini Pro, Gemini Flash).

The total-reward vectors are:

$$v_{\text{sync}} = (0.576, 0.525, 0.605, 0.589, 0.677, 0.512, 0.633, 0.471),$$

$$v_{\text{pancake}} = (0.805, 0.939, 0.865, 0.808, 0.767, 0.366, 0.350, 0.550),$$

$$v_{\text{interview}} = (0.792, 0.787, 0.696, 0.696, 0.739, 0.550, 0.837, 0.651).$$

For tasks a, b , compute centered cosine similarity:

$$\rho(a, b) = \frac{(v_a - \bar{v}_a) \cdot (v_b - \bar{v}_b)}{\|v_a - \bar{v}_a\|_2 \|v_b - \bar{v}_b\|_2},$$

and normalized angular distance:

$$d(a, b) = \frac{\arccos(\rho(a, b))}{\pi}.$$

The measured correlations are 0.162, 0.513, and 0.303; the corresponding distances are 0.448, 0.328, and 0.402. The average dispersion is:

$$D = \frac{2}{T(T-1)} \sum_{i < j} d(t_i, t_j) = 0.393, \quad T = 3.$$

This supports keeping all three tasks: they rank models differently, with the pancake task most separated and the sync/interview pair only moderately correlated. The metric is useful for data selection because redundant tasks would produce nearly identical reward vectors, while separated vectors indicate different supervision signals.

4 Discussion

The results are intentionally not a pure leaderboard of model intelligence. They also reflect agent interface maturity, tool-use choices, prompt following, media-domain heuristics, and verifier failure modes. GPT-5.4 did best on the pancake task, Claude Sonnet led the sync task, and Gemini Pro led the interview task. That spread is useful for RL benchmark design: the tasks expose different bottlenecks rather than rewarding only one style of agent behavior. The hard caps were important in practice because several runs looked superficially valid but failed a core requirement such as source authenticity, semantic preservation, or sync repair.

A CPU Sandbox Tool and Package List

Command-line tools: `ffmpeg`, `ffprobe`, `bash`, and `python >= 3.11`.

Python packages: `numpy`, `scipy`, `opencv-python`, `moviepy`, `av`, `librosa`, `soundfile`, `pydub`, `scenedetect`, `faster-whisper`, `Pillow`, `scikit-image`, `pandas`, `pydantic`, `tqdm`, `ultralytics`, `colour-science`, `noisereduce`, `imageio`, `imageio-ffmpeg`, `matplotlib`, and `jsonschema`.